

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 5

2 Computer Networking (awm22)

ISP ShinyNet connects customers to the Internet with a symmetric 40 Mbit/s link. The ISP gateway router/modem has a single buffer shared by all senders with a capacity of 1000 packets. You may assume all packets are of the same length and that packet-arrivals are not bursty.

A popular application uses three UDP flows each concurrently sharing the link. The three flows send at rates of 10 Mbit/s, 20 Mbit/s and 30 Mbit/s respectively.

- (a) What is the data rate successfully traversing the buffer and the fraction of packets for each flow dropped by the buffer? [3 marks]

Answer:

Approximately one third of packets will be lost by each flow traversing the link.

Flow 1 will achieve 6.67 Mbit/s

Flow 2 will achieve 13.33 Mbit/s

Flow 3 will achieve 20 Mbit/s

ShinyNet upgrades the customer gateway router/modem. The upgraded device employs one queue per flow (three queues in total for this case); each queue can hold a maximum number of packets in proportion to the flow count (333 packets per flow), the queues are scheduled using simple per-flow fair queueing with an equal weight for each queue.

- (b) What is the packet loss rate for each flow when using the upgraded gateway? [3 marks]

Answer:

The first flow loses 0% of its packets (the first flow consumes a share of the bandwidth equal to $\min(10 \text{ Mbit/s}, 40/3 \text{ Mbit/s}) = 10 \text{ Mbit/s}$)

The second and third flows equally share the remaining 3 Mbit/s of bandwidth, and therefore each get 1.5 Mbit/s. Hence, the second flow loses 25 and the third flow loses 50

- (c) Approximately how many packets does each flow have in its queue? [3 marks]

Answer: The queue for the first flow is empty. The queue for the second and third flow are at capacity (333 packets in each).

ShinyNet updates the gateway routers/switch to now use weighted-fair queueing. Using an unknown patented technology (or poor configuration); the following weights are assigned to the flows as follows: 0.2, 0.6, and 0.2 for each flow respectively.

- (d) What is the packet loss rate for each flow when using this gateway with weighted fair-queueing? [3 marks]

Answer:

In this configuration, the second flow consumes a bandwidth share of $\min(20 \text{ Mbit/s}, 0.6 \cdot 40 \text{ Mbit/s}) = 20 \text{ Mbit/s}$, and the first and third flows equally share the remaining 20 Mbit/s , i.e., each get 10 Mbit/s .

- (e) Approximately how many packets does each flow have in its queue? [3 marks]

Answer:

As a result, and assuming a queue capacity of 1000 packets queue, the first flow should lose close to 0%, though this system (dramatically bursty arrivals (significantly greater than 333 packets) may cause loss.)

The third flow loses approximately 67% of its packets.

The second flow's queue is empty or close to empty. The third flow's queue is about full with up to 333 packets. The first flow is allocated a fraction of the link bandwidth that matches its needs, its queue load is 1. If packets arrive regularly (are paced out, not in bursts), the queue for the first will remain 'essentially' empty.

- (f) Throughout this question we have assumed packets are of equal length; this is not a valid assumption for the vast majority of Internet traffic. Presuming that over the long term we wish for the queue discipline to retain the desired weighted-fairness properties, discuss the issues variable length packets raise, and propose an approach to maintain the desired weighted-fairness outcome. [5 marks]

Answer: We do not discuss generalised processor sharing and other logical extensions beyond WFQ. I would expect a less sophisticated but more example-driven response.

we have potential unfairness if we presume every packet is same length (WFQ as described requires this.)

To maintain WFQ behaviour at a packet level we need to count bytes (or bits) to attain the same level of weighted-fairness.

Until now we assume one packet is one credit, bit weighting means we now need to track the quantity consumed by each queue; as packets are of variable length we now need a credit mechanic incorporated into the scheduler. The credit mechanic will track the number of bits each packet consumed and permits appropriate adjustment of available scheduling resources to each (other) queue.
